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The critical points of entire fractions and B.Shapiro's 12th conjecture of entire functions

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Abstract

For any real polynomial $\boldsymbol{p(x)}$ of even degree $\boldsymbol{k}$. B. Shapiro propose the 12th conjecture saying that the sum of the number of real zeros of two polynomials $(\boldsymbol{(k-1)(p'(x))^2 - kp(x)p''(x)}$ and $\boldsymbol{p(x)})$ is larger than 0. We comprehensively prove the original B.Shapiro's 12th conjecture and B.Shapiro's 12th conjecture of entire functions by showing all cases that the conjecture are true, and the case that it is not true.

Keywords: entire fractions, critical points, polynomials, zeros

Mathematics Subject Classification: Primary 30C15 Secondary 30D20 , 26C10

1 Introduction

It is well known that the theory of entire functions began as a research field from the pioneer work of Laguerre[1], soon after the Weierstrass product representation became available. In 1900, in the survey of Borel[2]. It is stated that one of the core problems of the study of entire functions is finding relations between the zeros of a real entire function and the zeros of its derivatives. Here, an entire function is said to be real if it takes real values on the real axis.

Given a real polynomial $p(x)$, if $p(x)$ has all real and simple zeros, then the function $p(x)$ is (locally) strictly monotone was known to Gauss[3]. We can

reformulate it in the form of Laguerre inequality: Let $p(x)$ be a real polynomial with only real zeroes. Then $p(x)p^{''}(x)-[p^{'}(x)]^2 \leq 0$, $x \in \mathbb{R}$. The class of entire functions that are uniform limits on compact sets of sequences of polynomials with only real zeroes is the Laguerre-Pólya class (LP)[4]. And the entire functions of the Laguerre-Pólya class satisfy Laguerre inequality.

A refinement of the Laguerre inequality constitutes the Wiman conjecture and Pólya conjectures. The Wiman's conjecture was eventually confirmed by Bergweiler, Eremenko and Langley[6]. One Pólya's conjecture was settled by Craven, Csordas and Smith[9]. Another Pólya's conjecture was settled by Bergweiler, Eremenko[7].

By building upon results that resolved a conjecture of Wiman-Pólya's type conjectures, a number of conditions are established under which a real entire functions f must belong to the class LP. These conditions typically involve the differential polynomial in the form of $f(x)f^{''}(x)-\varkappa(f^{'}(x))^2$, where $\varkappa$ is a real number[5]. The zeroes of the function $f(x)f^{''}(x)-\varkappa(f^{'}(x))^2$ when f is a meromorphic function have been studied in [8].

Hawaiian conjecture was proposed when studying the Pólya's conjecture[9], saying that if $p(x)$ is a real polynomial, then the number of real zeroes of $(\frac{p^{'}(x)}{p(x)})^{'}$ does not exceed the number of non-real zeroes of $p(x)$. And the conjecture can be extended to entire functions[10]. The Hawaiian conjecture was proved in 2011 by Tyaglov[11]. Then, B. Shapiro propose three conjectures around the Hawaiian Conjecture(see Conjectures 11, 12 and 13 in [12]).

The 12th Conjecture by B. Shapiro which states: For any real polynomial $p(x)$ of even degree. $\sharp_r[(k-1)(p^{'}(x))^2-kp(x)p^{''}(x)]+\sharp_r p(x) > 0$. Here, k donates the degree of $p(x)$, $\sharp_r p(x)$ stands for the number of real zeros of a polynomial $p(x)$ with real coefficients.

In this paper, we extend the 12th Conjecture by B. Shapiro to entire functions as: For any real entire function $f(x)$. $\sharp_r[f(x)f^{''}(x)-\varkappa(f^{'}(x))^2]+\sharp_r f(x) > 0$. Here, $\varkappa$ is a real number. Especially, when $\varkappa = \frac{k-1}{k}$, k is the degree of polynomial $f(x)$, the B.Shapiro's 12th conjecture of entire functions is 12th conjecture of B.Shapiro.

Our results show that, in most cases, the conjecture of entire functions is true.

Theorem1 For any real entire function $f(x)$, when $f(x)$ has real zeros, or $f(x)$ has no real zeros and has at least one real critical point.

Then the quantity $\sharp_r[f(x)f^{''}(x)-\varkappa(f^{'}(x))^2]+\sharp_r f(x) > 0$ if and only if one of the following four cases holds: (1). $f(x)$ has real zeros; (2) $f(x)$ has at least two real critical points or one real critical point which its exponents is larger than 1; (3). $FF(+\infty) = 0$, or $FF(-\infty) = 0$. (4). $f(x)$ has one simple real critical point w, that is $f^{'}(x) = C(x)(x-w)$. Where $C(x)$ is a function with $C(w) \neq 0$, and the function $f(x)C^{'}(x)(x-w)+C(x)f(x)-\varkappa(C(x))^2(x-w)^2$ must has real zeros.

Theorem2 For any real entire function $f(x)$, when $f(x)$ has no real zeros and no real critical points.

Then the quantity $\sharp_r[f(x)f^{''}(x) - \varkappa(f^{'}(x))^2] + \sharp_r f(x) > 0$ if and only if one of the following two cases holds: (1). $FF(+\infty) = FF(-\infty) = A1$. $A1$ can be zero, constant and also can be the positive or negative infinity. (2). When $FF(x)$ has real critical points, $FF(+\infty) = 0$, $FF(-\infty) = A2$. Or, $FF(+\infty) = A2$, $FF(-\infty) = 0$, $A2$ can be a non-zero constant and also can be the positive or negative infinity.

For the rest two cases, B. Shapiro's 12th Conjecture of entire functions are false, they are Theorem 1.13 and Theorem 1.22. We also give examples to show cases described in Theorem 1, 2, 1.13 and 1.22 does occur.

2 The results

According to the classical theory of entire functions, the entire functions has three representations. If $f(s) \neq 0$, everywhere, then $f(s) = e^{P(s)}$. $P(s)$ is an entire functions. If there are finitely many zeros of f. Then $f(s) = (s-z_1)(s-z_2)\cdots(s-z_k)e^{P(s)}$. If there are infinitely many zeros of f. The zeros $z_1, z_2, \cdots$ of entire functions $f(s)$ of order ρ have the property. $\sum_{k=1}^{\infty} \frac{1}{(|z_k|)^{\rho+w}} < +\infty$, for all $w > 0$. Let p be the least integer $(p \leq \rho)$ such that $\sum_{k=1}^{\infty} |z_k|^{-p-1} < +\infty$. The following product representation holds (Hadamard's theorem on entire functions).

$f(s) = s^{\lambda} e^{P(s)} \prod_{k=1}^{\infty}(1 - \frac{s}{z_k}) \exp(\frac{s}{z_k} + \cdots + \frac{s^p}{pz_k^p})$.

Where $P(s)$ is a polynomial of degree not exceeding ρ. $\lambda = 0$ if $f(0) \neq 0$, and λ is the multiplicity of the zero $s = 0$ if $f(0) = 0$.

Let $G(s) = s^{\lambda} e^{P(s)}$, $G_{z_l}(s) = \exp(\frac{s}{z_k} + \cdots + \frac{s^p}{pz_k^p})$. $f(s) = G(s)\prod_{l=1}^{n}(1 - \frac{s}{z_l})G_{z_l}(s)$. $G(s)$, $G_{z_l}(s)$ have no zeros and poles. According to the representation of $G(s)$ and $G_{z_l}(s)$, we can obtain: $G^{'}(s) = \lambda s^{\lambda-1} e^{P(s)} + s^{\lambda} P^{'}(s) e^{P(s)}$, $G^{'}_{z_l}(s) = (\frac{1}{z_l} + \frac{s}{z_l^2} + \cdots + \frac{s^{p-1}}{z_l^p}) \exp(\frac{s}{z_l} + \cdots + \frac{s^p}{pz_l^p})$. $G^{'}(s)$ and $G^{'}_{z_l}(s)$ have no poles.

Lemma 2.1. *$f^{'}(s)$ has no poles.*

Proof For entire functions which have no any zeros and the finite number of zeros, their proofs which $p^{'}$ has no poles are all simple, here, we needn't to give. Observe that $\frac{f^{'}(s)}{f(s)} = \frac{G^{'}(s)}{G(s)} + \sum_{l=1}^{n}(\frac{G^{'}_{z_l}(s)}{G_{z_l}(s)} + \frac{(-\frac{1}{z_l})}{1-\frac{s}{z_l}}) = \frac{G^{'}(s)}{G(s)} + \sum_{l=1}^{n}(\frac{G^{'}_{z_l}(s)}{G_{z_l}(s)} + \frac{1}{s-z_l})$.

$f^{'}(s) = f(s)(\frac{G^{'}(s)}{G(s)} + \sum_{l=1}^{n}(\frac{G^{'}_{z_l}(s)}{G_{z_l}(s)} + \frac{1}{s-z_l})) = f(s)\frac{G^{'}(s)}{G(s)} + \sum_{l=1}^{n}(\frac{G^{'}_{z_l}(s)}{G_{z_l}(s)} f(s) + f(s)\frac{1}{s-z_l}) = G^{'}(s)\prod_{l=1}^{n}(1 - \frac{s}{z_l})G_{z_l}(s) + \sum_{l=1}^{n}(\frac{G^{'}_{z_l}(s)}{G_{z_l}(s)} G(s)\prod_{j=1}^{n}(1 - \frac{s}{z_j})G_{z_j}(s) + \frac{1}{s-z_l} G(s)\prod_{j=1}^{n}(1 - \frac{s}{z_j})G_{z_j}(s)) = G^{'}(s)\prod_{l=1}^{n}(1 - \frac{s}{z_l})G_{z_l}(s) + \sum_{l=1}^{n}(G^{'}_{z_l}(s)G(s)\prod_{j=1,j\neq l}^{n}(1 - \frac{s}{z_j})G_{z_j}(s) + \frac{1}{s-z_l} G(s)\prod_{j=1}^{n}(1 - \frac{s}{z_j})G_{z_j}(s))$.

$G(s)$ and $G_{z_l}(s)$ have no zeros and poles. $G^{'}(s)$ and $G^{'}_{z_l}(s)$ have no poles. So, $G^{'}(s)\prod_{l=1}^{n}(1 - \frac{s}{z_l})G_{z_l}(s)$ has no poles. $G^{'}_{z_l}(s)G(s)\prod_{j=1,j\neq l}^{n}(1 - \frac{s}{z_j})G_{z_j}(s)$ have no

poles. The factor $s - z_l$ in denominator is offset by the factor in $\prod_{j=1}^{n}(1 - \frac{s}{z_j})$, so, $\frac{1}{s-z_l}G(s)\prod_{j=1}^{n}(1 - \frac{s}{z_j})G_{z_j}(s)$ have no poles. So, $f^{'}(s)$ has no poles. □

Lemma 2.2. *For* $FF(s) = \frac{(f^{'}(s))^{\beta}}{f(s)}$. *Let* $\varkappa = \frac{1}{\beta}$, *the function* $\Delta 1 = f^{''}(s)f(s) - \varkappa(f^{'}(s))^2$. *When* $\varkappa > 1$, *all critical points of* $FF(x)$ *are all zeros of* $\Delta 1$. *When* $\varkappa = 1$, *all critical points of* $FF(s)$ *are all zeros of* $\Delta 1$. *When* $\varkappa < 1$, *all zeros of* $f^{'}(s)$ *are all critical points of* $FF(s)$. *Except zeros of* $f^{'}(s)$, *all critical points of* $FF(s)$ *are all zeros of function* $\Delta 1$.

Proof $FF^{'}(s) = (\frac{(f^{'}(s))^{\beta}}{f(x)})^{'} = \frac{\beta(f^{'}(s))^{\beta-1}f^{''}(s)f(s) - (f^{'}(s))^{\beta}f^{'}(s)}{(f(s))^2}$

$= \frac{\beta(f^{'}(s))^{\beta-1}(f^{''}(s)f(s) - \frac{1}{\beta}(f^{'}(s))^2)}{(f(s))^2}$.

$FF^{'}(s) = (\frac{(f^{'}(s))^{\beta}}{f(s)})^{'} = \frac{(f^{'}(s))^{\frac{1}{\varkappa}-1}(f^{''}(s)f(s) - \varkappa(f^{'}(s))^2)}{\varkappa(f(s))^2}$. When $\varkappa > 1$, zeros of $f^{'}(s)$ are all poles of $FF^{'}(s)$. All poles of $f^{'}(s)$ are all critical points of $FF(s)$. According to Lemma 1.1, $f^{'}(s)$ has no poles, so all critical points of $FF(s)$ are all zeros of function $\Delta 1$.

When $\varkappa = 1$, poles and zeros of $f^{'}(s)$ cannot be zeros and poles of $FF^{'}(s)$. All critical points of $FF(s)$ are all zeros of $\Delta 1$.

When $\varkappa < 1$, zeros of $f^{'}(s)$ are all critical points of $FF(s)$. So, except zeros of $f^{'}(s)$, all critical points of $FF(s)$ are all zeros of $\Delta 1$. □

Lemma 2.3. *If* $f(s)$ *has no real zeros. For* $f^{'}(s)$, *if its exponent of the real critical point* w *of* $f(s)$ *is larger than 1, then* w *is a real zero of* $\Delta 1$.

Proof According to the condition of this lemma, let $f^{'}(s) = C(s)(s-w)^l$, w is a real critical point of $f(s)$. And $C(w) \neq 0$, and it is a finite value. $f^{''}(s) = C^{'}(s)(s-w)^l + lC(s)(s-w)^{l-1}$.

$\Delta 1 = f(s)C^{'}(s)(s-w)^l + lC(s)f(s)(s-w)^{l-1} - \varkappa(C(s))^2(s-w)^{2l}$

$= (s-w)^{l-1}(f(s)C^{'}(s)(s-w) + lC(s)f(s)) - \varkappa(C(s))^2(s-w)^{2l}$.

$l > 1$, $f(w)C^{'}(w)(w-w) + lC(w)f(w) = lC(w)f(w) \neq 0$, and $lC(w)f(w)$ is a finite value. so, w is still a real zero of $\Delta 1$. □

Lemma 2.4. *If* $f(s)$ *has no real zeros and one simple real critical point* w, *that is* $f^{'}(s) = C(s)(s-w)$, *where* $C(s)$ *is a function with* $C(w) \neq 0$, *then* w *is not a zero of* $\Delta 1$.

Proof The w is critical point of $f(s)$. $C(w) \neq 0$, and it is a finite value. $f^{''}(s) = C^{'}(s)(s-w) + C(s)$.

$f(s)f^{''}(s) - \varkappa(f^{'}(s))^2 = f(x)C^{'}(s)(s-w) + C(s)f(s) - \varkappa(C(s))^2(s-w)^2$.

$f(w)C^{'}(w)(w-w) + C(w)f(w) - \varkappa(C(w))^2(w-w)^2 = C(w)f(w) \neq 0$. □

We derive Theorem 1 and Theorem 1.13 from a series of following lemmas.

Lemma 2.5. *When $f(x)$ has real zeros, where x is a real variable, let $\Delta 2 = f^{''}(x)f(x) - \varkappa(f^{'}(x))^2$. $\sharp_r \Delta 2 + \sharp_r f(x) > 0$.*

When $f(x)$ has no real zeros, $\sharp_r f(x) = 0$.

Lemma 2.6. *For $f(x)$ which has no real zeros, when $f^{'}(x)$ has at least 2 distinct real zeros, $\sharp_r \Delta 2 + \sharp_r f(x) > 0$.*

Proof The fraction $FF(x)$ is a real function. $f(x)$ has no real zeros, according to Lemma 1.1, $f^{'}(x)$ has no real poles. For $FF(x)$, there are no real poles, and it satisfies all conditions of Rolle's theorem, we can use Rolle's theorem.

According to the condition which $f^{'}(x)$ has at least 2 real zeros, so, between two adjacent real zeros of $FF(x)$, $FF(x)$ has at least one real critical point.

At least one real critical points of $FF(x)$ cannot be the zeros of $f^{'}(x)$, according to Lemma 1.2, at least one real critical point of $FF(x)$ is real zeros of function $\Delta 2$. So, $\sharp_r \Delta 2 \geq 1 > 0$. □

We give example 1.7 to show that case described in Lemma 1.6 does occur.

Example 2.7. $f_1(x) = e^{x^3 - 3x}$, $\varkappa$ can be any real number.

f_1 has no real zeros. $f_1^{'} = 3(x^2 - 1)e^{x^3 - 3x}$. $f_1^{'}$ has two distinct real zeros. In the interval $(-1, 1)$, there is one real zero of $f_1 f_1^{''} - \varkappa(f_1^{'})^2$. $\sharp_r[f_1 f_1^{''} - \varkappa(f_1^{'})^2] \geq 1 > 0$. And $\sharp_r f_1 = 0$.

Applying Lemma 1.3, obviously, we get Lemma 1.8.

Lemma 2.8. *When $f(x)$ has no real zeros and one real critical point. For $f^{'}(x)$, if the exponent of the critical point of $f(x)$ is larger than 1, then $\sharp_r \Delta 2 + \sharp_r f(x) > 0$.*

We give example 1.9 to show that case described in Lemma 1.8 does occur.

Example 2.9. $f_2(x) = e^{x^3}$, $\varkappa$ can be any real number.

f_2 has no real zeros, $f_2^{'} = 3x^2 e^{x^3}$. $f_2^{'}$ has one repeated zeros $w = 0$. For $f_2^{'}$, the exponent of the critical point $w = 0$ is 2 and larger than 1, so, $\sharp_r[f_2 f_2^{''} - \varkappa(f_2^{'})^2] \geq 1 > 0$. And $\sharp_r f_2 = 0$.

Lemma 2.10. *For $f(x)$ has no real zeros and one real simple critical point of $f(x)$. $FF(+\infty) = 0$, or $FF(-\infty) = 0$. $\sharp_r \Delta 2 + \sharp_r f(x) > 0$.*

Proof The fraction $FF(x)$ satisfies all conditions of Rolle's theorem. $f^{'}(x)$ has at least 2 real zeros, one is a finite zero of $FF(x)$, another is an infinite zero of $FF(x)$

on the real axis. So, between the finite zero and infinite of $FF(x)$ on the real axis, $FF(x)$ has at least one real critical point. So, $\sharp_r\Delta 2 \geq 1 > 0$. □

We give example 1.11 to show that case described in Lemma 1.10 does occur.

Example 2.11. $f_3(x) = e^{x^2}$, $\varkappa = 2$.

$f_3^{'} = 2xe^{x^2}$. $f_3^{'}$ has no real zeros and has one critical point $w = 0$, which exponent is 1. $f_3^{''} = 4x^2e^{x^2} + 2e^{x^2}$. $f_3f_3^{''} - \varkappa(f_3^{'})^2 = (2 - 4x^2)e^{x^2}$ has two real zeros. $FF_3 = \frac{\sqrt{2x}e^{\frac{x^2}{2}}}{e^{x^2}} = \sqrt{2x}e^{-\frac{x^2}{2}}$. $FF_3(+\infty) = FF_3(-\infty) = 0$. So, $\sharp_r[f_3f_3^{''} - \varkappa(f_3^{'})^2] \geq 2 > 0$. And $\sharp_r f_3 = 0$.

Lemma 2.12. *For $f(x)$ has no real zeros and one real simple critical point, that is $f^{'}(x) = C(x)(x - w)$, where $C(x)$ is a function with $C(w) \neq 0$. $FF(+\infty) \neq 0$ and $FF(-\infty) \neq 0$. The function $f(x)C^{'}(x)(x-w)+C(x)f(x)-\varkappa(C(x))^2(x-w)^2$ has real zeros, $\sharp_r\Delta 2 + \sharp_r f(x) > 0$.*

Proof In the proof of Lemma 1.4, we have: $f(x)f^{''}(x) - \varkappa(f^{'}(x))^2 = f(x)C^{'}(x)(x - w)+C(x)f(x)-\varkappa(C(x))^2(x-w)^2$. When $f(x)C^{'}(x)(x-w)+C(x)f(x)-\varkappa(C(x))^2(x-w)^2$ has real zeros, $\sharp_r[f(x)f^{''}(x) - \varkappa(f^{'}(x))^2] > 0$. So, $\sharp_r\Delta 2 + \sharp_r f(x) > 0$. □

For the case which $f(x)$ has real zeros, or $f(x)$ has no real zeros and has at least one real critical point, except five cases of Lemma 1.5, Lemma 1.6, Lemma 1.8, Lemma 1.10 and Lemma 1.12, only one case is remained. According to the result and proof of Lemma 1.12, under this case, B.Shapiro's 12th conjecture of entire functions is not true. So, we have:

Theorem 2.13. *For any real entire function $f(x)$, when $f(x)$ has no real zeros and has at least one real simple critical point.*

Then the quantity $\sharp_r[f(x)f^{''}(x) - \varkappa(f^{'}(x))^2] + \sharp_r f(x) = 0$ if and only if $FF(+\infty) \neq 0$ and $FF(-\infty) \neq 0$; $f^{'}(x) = C(x)(x - w)$, where $C(x)$ is a function with $C(w) \neq 0$, and the function $f(x)C^{'}(x)(x - w) + C(x)f(x) - \varkappa(C(x))^2(x - w)^2$ has no real zeros.

We give example 1.14 to show that case described in Theorem 1.13 does occur.

Example 2.14. $f_4(x) = f_3(x) = e^{x^2}$, $\varkappa = 1$.

$f_4f_4^{''} - \varkappa(f_4^{'})^2 = 2e^{2x^2}$ has no real zeros. $FF_4 = \frac{2xe^{x^2}}{e^{x^2}} = 2x$. $FF_4(+\infty) = +\infty$, $FF_4(-\infty) = -\infty$. So, $\sharp_r[f_4f_4^{''} - \varkappa(f_4^{'})^2] = 0$, and $\sharp_r f_4 = 0$. It is contrast to Shapiro's conjecture.

Applying Lemma 1.5, Lemma 1.6, Lemma 1.8, Lemma 1.10, Lemma 1.12 and Theorem 1.13, we get Theorem 1.

When $p(x)$ is a polynomial of even degree, $p(x)$ has real zeros, or $p(x)$ has no real zeros and has at least one real critical point. The $p(x)$ satisfies the condition of Theorem 1.13 and Theorem 1, let $\varkappa = \frac{k-1}{k}$, k stands for degree of polynomial $p(x)$, the 12th conjecture of B.Shapiro of entire functions is the original 12th conjecture of B.Shapiro. So, the original 12th conjecture of B.Shapiro is also proved.

When the entire functions $f(x)$ has no real zeros and no real critical points. If $FF(+\infty)$ and $FF(-\infty)$ obtain the positive and negative values with the different positive and negative symbol, the positive and negative values can contain the positive or negative infinity. According to that $FF(x)$ is continuous on the real axis, on the real axis, there is at least one zero of $FF(x)$. In the front, we have already given all results which $FF(x)$ has at least one real zero. So, this case belongs to the situation which we have already given in front.

We give an example that $f(x)$ has no real zeros and no real critical points. $f(x) = (x^2 + x + 4)e^x$. $f^{'}(x) = (x^2 + 3x + 5)e^x$. Obviously, neither $f(x)$ nor $f^{'}(x)$ has real zeros. So, in the following, we still need to give the results which $f(x)$ has no real zeros and no real critical points.

We derive Theorem 2 and Theorem 1.22 from a series of following lemmas.

Lemma 2.15. *For $f(x)$ has no real zeros and no real critical points, if $FF(+\infty) = FF(-\infty) = 0$, then $\sharp_r \Delta 2 + \sharp_r f(x) > 0$.*

Proof Now suppose $FF(+\infty) = FF(-\infty) = 0$, according to Rolle's Theorem, on the real axis, there is at least one critical point. At least one critical point is the zero of function $\Delta 2$. $\sharp_r \Delta 2 \geq 1 > 0$. $\sharp_r f(x) = 0$. □

We give example 1.16 to show that case described in Lemma 1.15 does occur.

Example 2.16. $f_5(x) = e^{-e^x}$, $\varkappa < 1$.

$f_5^{'} = -e^{-e^x+x}$. f_5 has no real zeros and critical points.

$FF_5 = \frac{(f_5^{'})^{\frac{1}{\varkappa}}}{f_5} = (-1)^{\frac{1}{\varkappa}} e^{(1-\frac{1}{\varkappa})e^x + \frac{x}{\varkappa}}$. When $(1 - \frac{1}{\varkappa}) < 0$, $\varkappa < 1$, when $x = +\infty$, $FF_5(+\infty) = 0$. $x = -\infty$, $FF_5(-\infty) = 0$. $\sharp_r[f_5 f_5^{''} - \varkappa(f_5^{'})^2] + \sharp_r f_5 > 0$.

Lemma 2.17. *For $f(x)$ has no real zeros and no real critical points, if $FF(+\infty) = FF(-\infty) = \infty$, ∞ is the infinity which has the same positive and negative symbol, then $\sharp_r \Delta 2 + \sharp_r f(x) > 0$.*

Proof $(\frac{1}{FF(x)})^{'} = -\frac{FF^{'}(x)}{FF^{2}(x)}$. So, $FF(x)$ and $\frac{1}{FF(x)}$ have the same critical points. $\frac{1}{FF(+\infty)} = \frac{1}{FF(+\infty)} = 0$. Repeat the proof of Lemma 1.15, we can obtain $\frac{1}{FF(x)}$ has at least one real critical point, so, $FF(x)$ has at least one real critical point. □

We give example 1.18 to show that case described in Lemma 1.17 does occur.

Example 2.18. $f_6(x) = e^{x^3+3x}$, $\varkappa = 1$.

$f_6^{'} = (3x^2+3)e^{x^3+3x}$. f_6 has no real zeros and critical points. $FF_6 = \frac{f_6^{'}}{f_6} = 3x^2+1$. $FF_6(+\infty) = FF_6(-\infty) = +\infty$.

Lemma 2.19. *For $f(x)$ has no real zeros and no real critical points, if $FF(+\infty) = FF(-\infty) = A$, A is the constant which has the same positive and negative symbol, then $\sharp_r\Delta 2 + \sharp_r f(x) > 0$.*

Proof Applying Rolle's Theorem, $FF(x)$ has at least one real critical point. □

We give example 1.20 to show that case described in Lemma 1.19 does occur.

Example 2.20. $f_7(x) = (x^2+20)e^x$, $\varkappa = 1$.

$f_7^{'} = (x^2+2x+10)e^x$. f_7 has no real zeros and critical points. $FF_7 = \frac{f_7^{'}}{f_7} = \frac{x^2+2x+20}{x^2+20}$. $FF_7(+\infty) = FF_7(-\infty) = +1$.

Lemma 2.21. *For $f(x)$ has no real zeros and no real critical points, if $FF(+\infty) = 0$, $FF(-\infty) = A$. Conversely, If $FF(+\infty) = A$, $FF(-\infty) = 0$, A can be a non-zero constant and it also can be the positive or negative infinity. When $FF(x)$ has the real critical points, $\sharp_r\Delta 2 + \sharp_r f(x) > 0$.*

The result of Lemma 1.21 is obvious, we needn't to give its proof.

For the case which $f(x)$ has real zeros and no real critical points, except four cases of Lemma 1.15, Lemma 1.17, Lemma 1.19 and Lemma 1.21, only one case is remained, under this case, 12th conjecture of B.Shapiro of entire functions is not true. So, we have:

Theorem 2.22. *For any real entire function $f(x)$, when $f(x)$ has no real zeros and no real critical points.*

Then the quantity $\sharp_r[f(x)f^{''}(x) - \varkappa(f^{'}(x))^2] + \sharp_r f(x) = 0$ if and only if $FF(x)$ has no any real critical points; $FF(+\infty) = 0$, $FF(-\infty) = A$. Or, $FF(+\infty) = A$, $FF(-\infty) = 0$, A can be a non-zero constant and also can be the positive or negative infinity.

We give example 1.23 to show that case described in Theorem 1.22 does occur.

Example 2.23. $f_8(x) = e^x$, $\varkappa > 1$.

$f_8' = e^x$. f_8 has no real zeros and critical points. $FF_8 = \frac{(f_8')^{\frac{1}{\varkappa}}}{f_8} = \frac{e^{\frac{x}{\varkappa}}}{e^x} = e^{(\frac{1}{\varkappa}-1)x}$. $f_8'' = e^x$. $f_8 f_8'' - \varkappa(f_8')^2 = (1-\varkappa)e^{2x}$. $\varkappa > 1$. $FF_8(+\infty) = 0$. $FF_8(-\infty) = +\infty$. $\varkappa < 1$. $FF_7(+\infty) = +\infty$. $FF_7(-\infty) = 0$. It is contrast to Shapiro's conjecture.

Applying Lemma 1.15, Lemma 1.17, Lemma 1.19, Lemma 1.21 and Theorem 1.22 we get Theorem 2.

Declarations

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